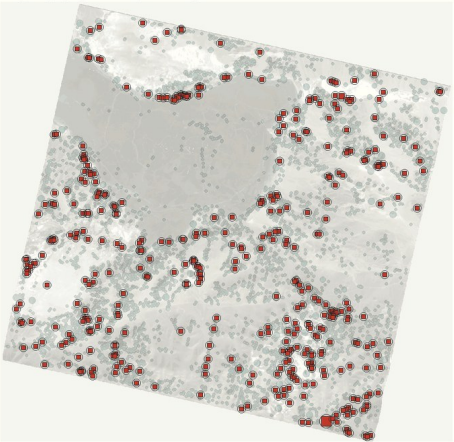
Allotrope v1.0

Hyperspectral anomaly detection and material identification

Any sensor in, georeferenced answers out:
a boolean anomaly map, objects called man-made or natural, and the closest material class.

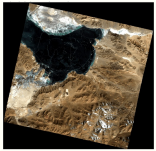
AIGC Stage-III evaluation
Team Allotrope · 4 November 2026, 09:00



PRISMA, 14 Feb 2025: 405 man-made objects of 5,544 anomalies.

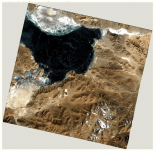
0.189	449 / 909	+0.17
pAUC, false alarms ≤1 %	objects in the top 1 %	man-made recall over ACE
HyperSL: 0.095	HyperSL: 369	PRISMA, quarter-pixel targets

Scene to materials



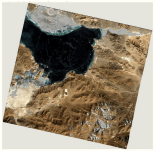
1 Scene

1237×1202 px, 239 bands



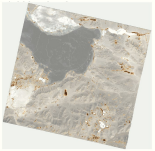
2 Band filter

186 of 239 bands usable; 30.3 % no-data



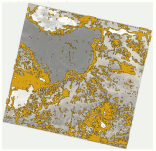
3 Segmentation

thick cloud left out, 4.9 %



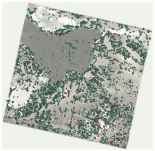
4 Score

Indradhanu V2 scores every pixel



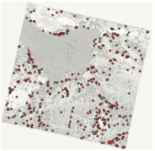
5 Threshold

top 8 %: 118,950 pixels



6 Namer

5,544 objects



7 Selection

405 man-made

One PRISMA scene, 1.5 million pixels, narrowed in seven recorded steps to the objects an analyst asks for.

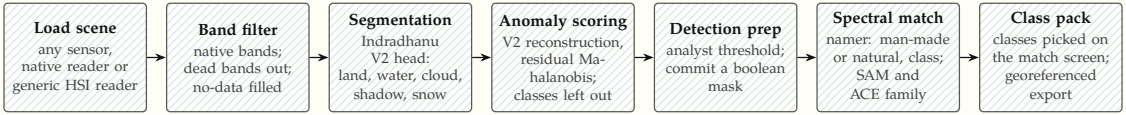
THE SAME CHAIN ON THREE SENSORS

	Tanager	PRISMA	AVIRIS-NG
anomaly objects	1,103	5,544	2,375
man-made at $P(\mathrm{M}) > 0.5$	272	405	1,290
man-made at 0.25 (very sensitive)	585	2,263	1,716
oil/fuel + plastic	247	141	936

WHAT THE EVALUATORS RECEIVE PER SCENE

Map: boolean anomaly mask, GeoTIFF on the scene's UTM grid
Vectors: pixel and object shapefiles with call, class and SAM match
Table: one row per object, with latitude and longitude
Reflectance: L2 surface reflectance from L1, GeoTIFF and BSQ
No heatmap in any submission pack

Pipeline



Seven actions in one dependency graph; each step writes what the next reads, and the export restores the scene's real map grid.

TIME PER PRISMA SCENE, CPU, 16 THREADS

V2 reconstruction, two passes	402 s
V2 segmentation	47 s
residual Mahalanobis	4 s
V2 total (FT04: 274 s)	**454 s**
namer, PRISMA / Tanager	9.2 / 7.6 s
class-pack export	2.1–2.6 s

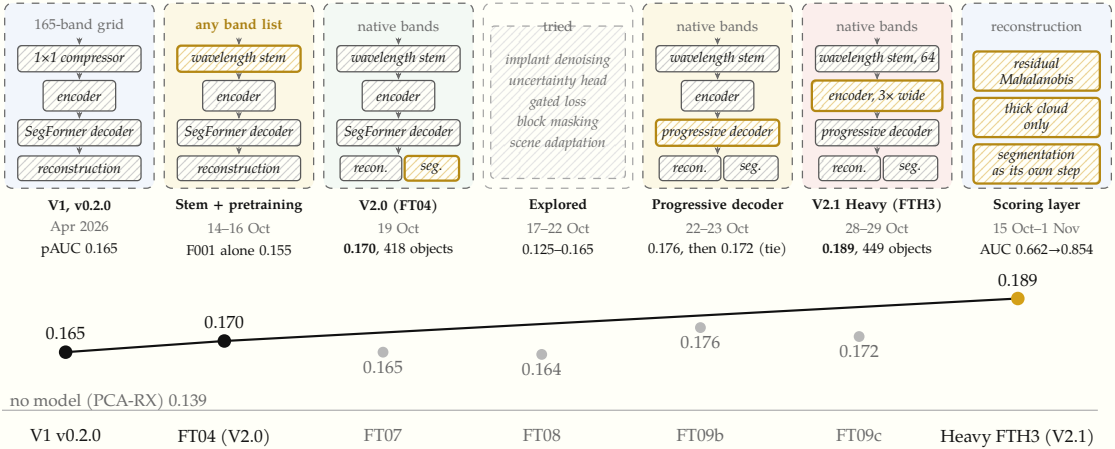
HOW IT RUNS

- The job queue is a Postgres table with row locking and heartbeats; no message broker.
- A web API and one worker; the browser polls. One origin behind nginx.
- Science code never touches the database.

SENSORS END TO END

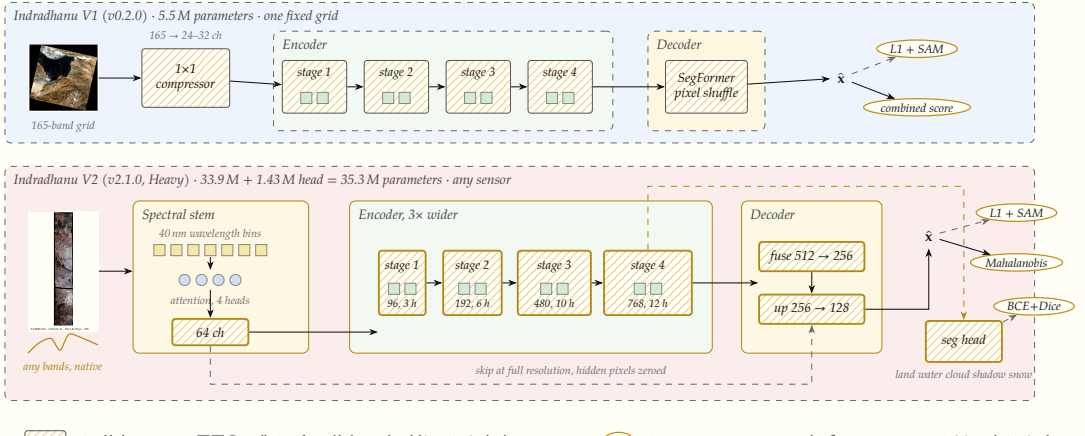
hyperspectral	PRISMA, EnMAP, AVIRIS-NG, Tanager
any HSI file	GeoTIFF, ENVI, HDF5, NetCDF (GHOSt is the first user)
thermal	Landsat-9, HotSat-1 (Chakshu)

Architecture evolution



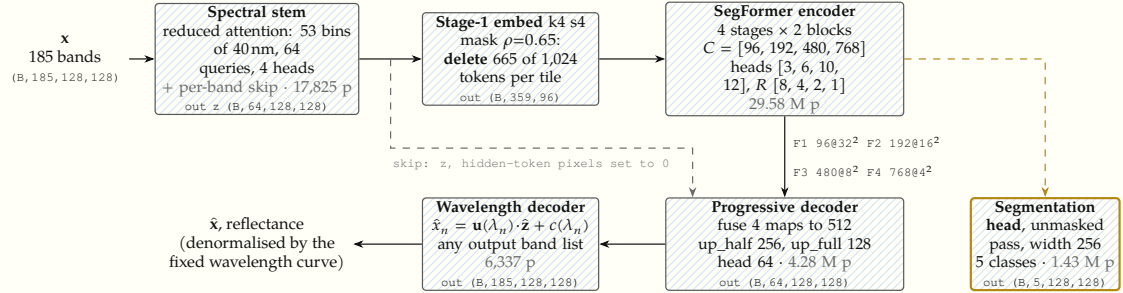
Shipped generations in ink, experiments that did not beat the shipped model in grey; each column adds the part outlined in ochre.

Indradhanu V1 and V2



V1 v0.2.0 0.165 → V2 FT04 0.170 → V2 Heavy 0.189 pAUC on the labelled PRISMA bench, cloud excluded; 418 → 449 of 909 objects in the top 1 %.

Indradhanu V2



OBJECTIVE, ON HIDDEN PIXELS M ONLY

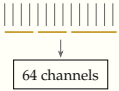
$$\mathcal{L} = \underbrace{\frac{1}{C|M|}\sum_{p\in M}\sum_{c}|\hat{x}_{c,p}-x_{c,p}|}_{L_1} + \lambda(t)\ \underbrace{\frac{1}{|M|}\sum_{p\in M}\angle(\mathbf{x}_p,\hat{\mathbf{x}}_p)}_{L_{\mathrm{SAM}}} + 0.1\,\mathcal{L}_{\mathrm{seg}}$$

$\lambda(t) = 0.5\min(1, t/10)$; $\mathcal{L}_{\mathrm{seg}}$ = BCE + Dice. A pixel is always scored from the pass in which it was hidden.

AS TRAINED

	pretraining (F002)	fine-tuning (FTH3)
data	SpectralEarth	EnMAP, PRISMA, AVIRIS-NG
epochs	80	18 + 6
learning rate, cosine	3×10^{-4}	3×10^{-5}
batch	32, bf16, 1 H100	16 × 2, bf16
mask ratio	0.65	0.65
validation SAM	3.89°	PRISMA 4.48°

Components



Wavelength-aware spectral stem

SHIPPED

Attention over 40 nm wavelength bins (53 bins, 64 queries, 4 heads) plus a per-band skip path into 64 channels. Takes any band list; 17,825 parameters.

Effect. About 6× faster than per-band attention at equal detail; reads PRISMA's 185 native bands without resampling.



Band-list augmentation

SHIPPED

80 % of training batches read a random band subset (50–95 % thinning, 1–3 gaps of 20–150 nm, or a 6–15 nm grid; never under 40 bands); stated wavelengths jittered ±3 nm.

Effect. Fixes the failure it was built for: before it, native PRISMA Set-1 scored ROC-AUC 0.205 against 0.725 when resampled.

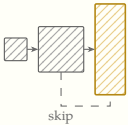


Segmentation head

SHIPPED

An unmasked pass through the same encoder into a width-256 head: land, water, cloud, thin cloud, shadow, snow. 1.43 M parameters.

Effect. IoU water 0.825, cloud 0.749. Excluding thick cloud only lifts pAUC 0.175 → 0.189.



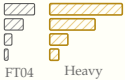
Progressive decoder

SHIPPED

Fuse the four encoder maps to 512, then up-sample 256 → 128 with a full-resolution skip from the stem, hidden pixels zeroed.

Effect. Alone (FT09b): +0.0060 [+0.0002, +0.0118] pAUC, then a tie five epochs later. Kept in Heavy, pretrained from the start.

Components, continued



Heavy scaling

SHIPPED

Encoder widths ×3, [96, 192, 480, 768], heads [3, 6, 10, 12]; spectral width 64; 33.9 M parameters. 80 epochs on one H100.

Effect. Pretrained alone 0.166 vs 0.155; fine-tuned 0.189 vs FT04's 0.170, +0.0196 [+0.0072, +0.0316].



Residual Mahalanobis score

SHIPPED

Whiten the reconstruction residual in its top 30 directions; measure each pixel against the mean of a ring (21/61 px) of neighbours.

Effect. Mean ROC-AUC 0.662 → 0.854 on the same model; 3–4 s per scene.

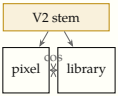


Uncertainty head, gated training

TRIED, NOT SHIPPED

A per-pixel scale head (FT06); a gated loss (FT07) and block masking (FT08); implant denoising (FT05); per-scene adaptation.

Effect. Uncertainty 0.125 / 0.163, gated loss 0.165, block masking 0.164, adaptation ≈0.160, all against 0.170. None shipped.



Spectral namer

SHIPPED

Two towers on V2's stem and normalisation (17,825 of 347,685 parameters) match a pixel against library spectra.

Effect. Man-made recall at ACE's false-alarm rate +0.14 to +0.20 over ACE at quarter-pixel fill.

Anomaly score

PER PIXEL p

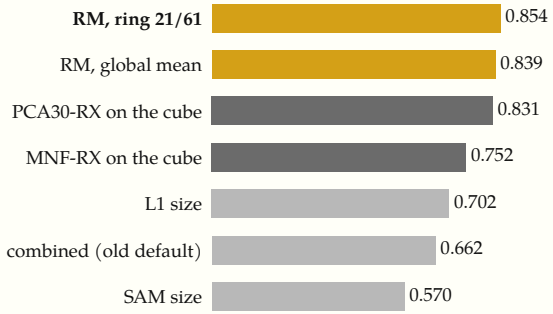
$$\mathbf{r}_p = \mathbf{x}_p - \hat{\mathbf{x}}_p \in \mathbb{R}^C$$

$$\mathbf{z}_p = \Lambda_d^{-1/2} \mathbf{U}_d^\top (\mathbf{r}_p - \mu_r), \quad d = 30$$

$$s(p) = \left\| \mathbf{z}_p - \frac{1}{|\mathcal{R}_p|} \sum_{q \in \mathcal{R}_p} \mathbf{z}_q \right\|^2$$

- $\mathbf{U}_d, \Lambda_d$: top 30 eigenpairs of the residual covariance over up to 200,000 valid pixels. Stabilised by truncation; 465 covariance entries instead of 17,205 for 185 bands.
- $\mathcal{R}_p$: the 3,280-pixel ring between a 21×21 guard and a 61×61 window, summed with integral images in four look-ups.
- Plain RX over all bands was dropped: near-singular covariance.

MEAN ROC-AUC, FOUR PRISMA SCENES, SAME V1 RECONSTRUCTION

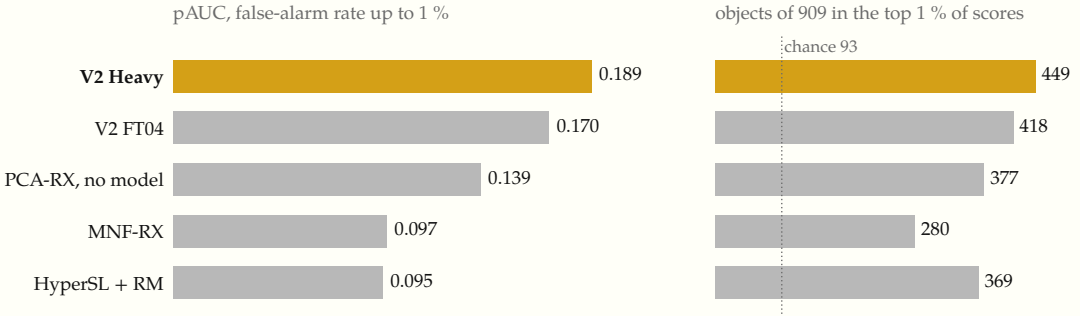


Scoring the residual's direction, not its size: 0.662 → 0.854.

The 21/61 window was the best of four on these same scenes; PCA30-RX still wins Set-3 (0.766 vs 0.749).

Detection results

LABELLED PRISMA BENCH: FOUR SCENES, 909 OBJECTS, THICK CLOUD EXCLUDED



Indradhanu V2 Heavy finds 449 of 909 labelled objects in the top 1 % of its scores; the no-model baseline finds 377.

AGAINST FT04 (PAIRED BOOTSTRAP)

pAUC **+0.0196** [+0.0072, +0.0316]; $P(B > A) = 1.000$. Resolved at three neighbouring checkpoints.

BY DIFFICULTY, TOP 1 %

easy 58.2 → **62.2** %; medium 39.3 → **43.9** %; subtle 40.3 → **42.0** %.

PRODUCT = EVALUATOR

The shipped path reproduces the evaluator's per-scene pAUC to four decimals: 0.2617, 0.1874, 0.0861, 0.2217.

Detection detail

ALL MEASURES, 909 OBJECTS

model	pAUC	top 1 %	top 0.1 %	ROC	$TPR_{10^{-3}}$
FT04	0.162	45.7 %	14.9 %	0.853	0.061
FT04, cloud excl.	0.170	46.0 %	15.8 %	0.852	0.065
FTH2 ep 8, thick excl.	0.186	49.0 %	15.7 %	0.870	0.066
Heavy FTH3 ep 5	0.175	46.4 %	13.9 %	0.866	0.058
Heavy FTH3 ep 5, thick excl.	**0.189**	**49.4 %**	**16.0 %**	**0.869**	**0.065**
no model (PCA-RX)	0.128	39.7 %	8.1 %	0.839	0.029
MNF-RX	0.075	27.4 %	3.4 %	0.749	0.019
chance		10.2 %	1.1 %		

PER SCENE, HEAVY FTH3 EPOCH 5

	Set-1	Set-2	Set-3	Set-4
thick cloud excluded	0.00 %	0.00 %	1.49 %	0.47 %
pAUC	0.2617	0.1874	0.0861	0.2217
ROC-AUC	0.9357	0.8738	0.7807	0.8877

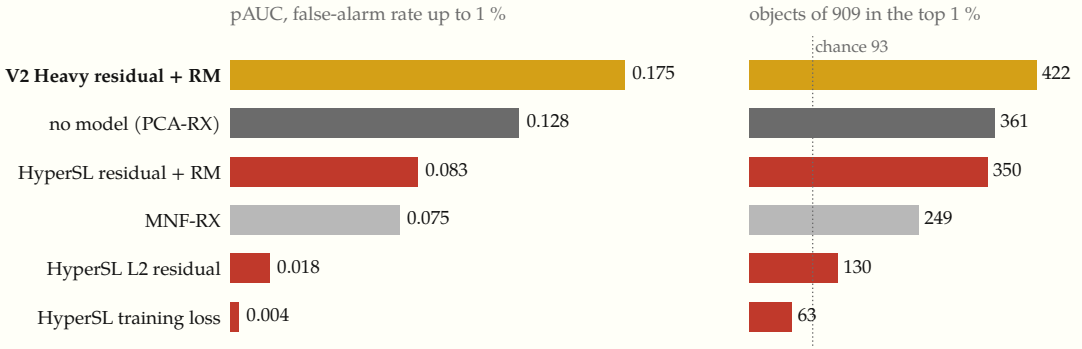
PAIRED BOOTSTRAP AGAINST FT04

epoch	pAUC difference	top 0.1 %
4	+0.0148 [+0.0029, +0.0259]	−0.4 pts
5	**+0.0196 [+0.0072, +0.0316]**	+0.1 pts
6	+0.0179 [+0.0055, +0.0292]	−0.3 pts

- Objects, not pixels, are resampled: 909 with replacement, both models on each draw.
- The top 0.1 % is a tie; the gain is in the broad ranking.
- Three neighbouring checkpoints all resolve, so the gain is not checkpoint luck.

Against HyperSL

SAME BENCH, SAME SCORER, SAME PIXELS; HYPERSL'S PUBLISHED CHECKPOINT AND CODE UNMODIFIED



HyperSL's reconstruction makes detection worse than using none: 0.083 against 0.128 with no model at all.

paired difference [95 %]	pAUC	top 1 % (pts)		pAUC	top 1 % (pts)
HyperSL − no model	−0.045 [−0.059, −0.032]	−1.2 [−4.0, +1.5]	no model − V2	−0.047 [−0.060, −0.033]	−6.7 [−8.8, −4.5]
HyperSL − V2	−0.092 [−0.106, −0.078]	−7.9 [−10.8, −5.2]	same, cloud excluded	−0.094 [−0.109, −0.080]	−8.8 [−11.7, −6.2]

Why not HyperSL

1 A PIXEL REBUILDS ITSELF

HyperSL hides bands within a pixel and rebuilds them from that pixel's own visible bands, so an anomaly reconstructs itself. Its own training loss as a score catches 63 objects, below the ≈93 of random scores; ROC-AUC 0.501.

2 PER-SCENE MIN-MAX SCALING

$$x'_b = \frac{x_b - m_b}{M_b - m_b}, \qquad m_b, M_b \text{ from the scene}$$

The same spectrum $(0.20, 0.50)$ becomes $(0.5, 1.0)$ in one scene and $(0.43, 0.75)$ in another; one bright anomaly rescales every pixel; an angle of 15.07° becomes 36.87°. A library spectrum has no scene: 41 % of ECOSTRESS values clipped, and a spectrum found its own resampled copy only 28 % of the time.

3 CLOUD-FREE PRETRAINING

Same classifier on EnMAP's cloud labels: cirrus recall **28.9 %** from HyperSL features against **67.9 %** from the raw bands. In its favour, thick cloud 87.1 % vs 80.4 %.

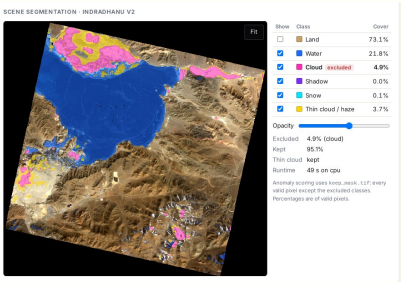
4 A SMOOTHING UNMIXING HEAD

HySU target side (true area, px)	0.25 m (0.13)	0.5 m (0.51)	1 m (2.0)
ACE, library + field spectra	**0.97**	**1.00**	1.00
HyperSL, anchored	0.68	0.77	1.00

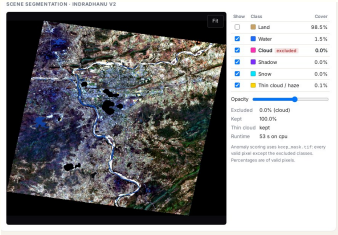
THE FIXES IT WOULD NEED

A fixed normalisation curve, wavelength tokens, whole hidden pixels, clouds in pretraining: the design Indradhanu already has. Not adopted.

Cloud masking



PRISMA, V2 segmentation in the product: land 73.1, water 21.8, cloud 4.9 (left out), thin cloud or haze 3.7, snow 0.1 %.



EnMAP, the same step and head.

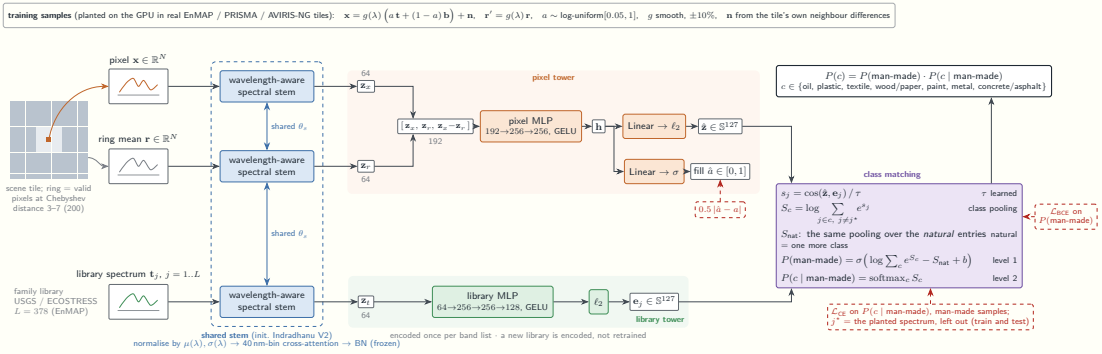
THE HEAD, VALIDATION

IoU water	0.825
IoU cloud	0.749
IoU thin cloud	0.090
rare classes weighted	×3

THE RULE

- Heavy's thin-cloud output fires on haze: 75.4 % of Set-4's pixels and 209 of its 230 targets.
- So only thick cloud is left out: pAUC 0.175 → 0.189.
- Water stays in: vessels are targets.

Spectral namer



Pixel tower and library tower share Indradhanu V2's stem; level 1 (man-made or natural) and level 2 (the class) are both read from the same matching scores.

Namer model

TWO LEVELS, ONE SET OF SCORES

$$
\begin{aligned}
s_j &= \cos(\mathbf{z}, \mathbf{e}_j)/\tau && \text{library entry } j \\
S_c &= \log \textstyle\sum_{j \in c} e^{s_j} && \text{class score} \\
\ell &= \mathrm{LSE}(S_0, \dots, S_6) - S_{\text{nat}} + b && \text{level 1} \\
P(c) &= \sigma(\ell) \cdot \mathrm{softmax}_c(S_c) && \text{level 2}
\end{aligned}
$$

$P(\text{man-made}) = \sigma(\ell)$ is the share of the pixel's matching mass on man-made library spectra. τ is learned. A new library spectrum needs no retraining.

CLASSES

oil/fuel, plastic, textile/fabric, wood/paper, paint/coatings, metal, concrete/asphalt; natural spectra form the natural shelf.

TRAINED ON PLANTED PIXELS FROM REAL TILES OF THREE SENSORS

$$\mathbf{x} = \mathbf{g} \odot (a\,\mathbf{t} + (1-a)\,\mathbf{b}) + \mathbf{n}, \quad a \sim \text{log-uniform}[0.05, 1]$$

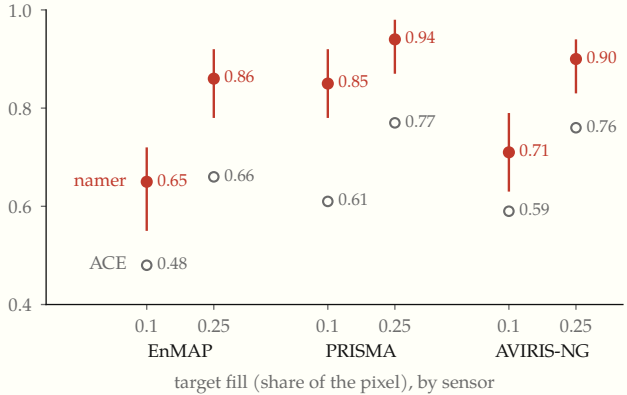
t a library spectrum (50 % man-made, 25 % natural, 25 % none), **b** the real background, **g** a smooth gain within ±10 %, **n** the tile's own noise.

$$\mathcal{L} = \mathrm{BCE}(\ell, \mathrm{M}) + \mathrm{CE}(S_{0..6}, c) + 0.5\,|\hat{a} - a|$$

library spectra	378 / 338 / 380 (EnMAP / PRISMA / AVIRIS-NG)
held out	20 % of each man-made class
parameters	347,685, of which the shared stem 17,825
optimiser	AdamW 3×10^{-4}, stem at 0.3×
run	40 epochs × 500 steps on one H100

Namer against ACE

LEVEL 1: MAN-MADE RECALL AT ACE'S OWN NATURAL FALSE-ALARM RATE



At quarter-pixel fill the namer finds 86–94 % of man-made targets; ACE 66–77 %.
Bars: 95 % bootstrap over library spectra (221–254 per sensor, 500 draws). ACE: leave-one-spectrum-out.

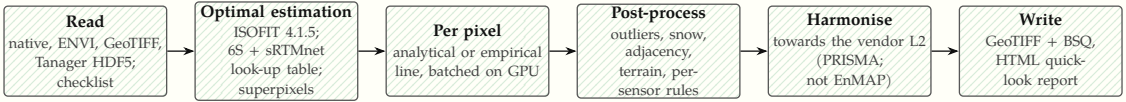
LEVEL 2: CLASS ACCURACY ON UNSEEN SPECTRA

sensor, fill	namer [95 %]	ACE
EnMAP 0.1	**0.59** [0.48, 0.70]	0.24
EnMAP 0.25	**0.68** [0.55, 0.79]	0.47
PRISMA 0.1	**0.69** [0.53, 0.81]	0.41
PRISMA 0.25	0.70 [0.54, 0.82]	0.60
AVIRIS-NG 0.1	**0.65** [0.53, 0.76]	0.38
AVIRIS-NG 0.25	0.70 [0.57, 0.82]	0.58

THE OPEN SET (TEXTILE HELD OUT)

- Unseen textile: called man-made 46–47 %; then a confident wrong class 45–46 %.
- Abstention sends only 1–4 % to "unknown".
- On known classes the flag works: 0.80 / 0.84 / 0.81 accurate at 90 % coverage. The product says "closest known class".

Atmospheric correction



One offline container turns L1 radiance into L2 surface reflectance for EnMAP, PRISMA, AVIRIS-NG, Tanager and GHOSt.

END-TO-END TRIALS OF THE SHIPPED CONTAINER

scene	reference	time	land MAE
PRISMA, Maharashtra coast	ASI L2C	33 min	0.0124
Tanager, Thessaloniki	Planet SR	36 min	0.0114
Tanager, ROCX2025	Planet SR	44 min	0.0089
PRISMA, Himachal, 18.5 % snow	ASI L2D	26 min	0.0461
EnMAP, Sundarbans, haze	DLR L2A	33 min	0.0157
Tanager, Permian, gas flares	Planet SR	36 min	0.0092
AVIRIS-NG full line, 4.0 Mpix	JPL	**41 min**	**0.0057**

HELD-OUT ACCURACY

- PRISMA, five unseen scenes against ASI: land MAE 0.0244 → **0.0099** after harmonisation.
- EnMAP against DLR: 0.0098 already; harmonisation did not help, so it is off.
- AVIRIS-NG against JPL: bias +0.0031, median spectral angle 2.27°.

THE PRODUCT

Reflectance ×10,000 as 16-bit integers, no-data −9999, water-vapour bands dropped and listed, wavelengths and FWHM in both files; outlier, snow and provenance sidecars.

Any sensor

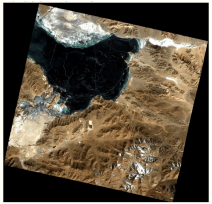
sensor	input	map grid from	checked
PRISMA	L2D HE5, native bands	the file's UTM corners	ground-truth grid reproduced exactly
EnMAP	L2A, 224 bands	vendor GeoTIFF	segmentation and scoring end to end
AVIRIS-NG	ENVI, 372 bands	ENVI map info	2,375 objects on the match screen
Tanager	HDF5 or ENVI; 426 bands, 368 good	HDF-EOS grid	UTM 13N, 30 m
any HSI file	GeoTIFF, ENVI, HDF5, NetCDF	file CRS or geolocation file	pre-flight report: bands, FWHM, scale, no-data

THE PRISMA GEOREFERENCE FIX

$$\mathbf{A} = \begin{pmatrix} d_x & 0 & UL_e - \frac{d_x}{2} \\ 0 & -d_y & UL_n + \frac{d_y}{2} \end{pmatrix}, \quad d_x = \frac{UR_e - UL_e}{W - 1}, \; d_y = \frac{UL_n - LL_n}{H - 1}$$

The corners are pixel centres. The old latitude/longitude box fit was a median **2.4 px** off (95th percentile 4.1, max 5.6). Internal evaluation was never affected: it works in pixel space.

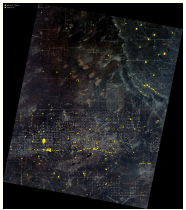
V2 reads every band within 15 nm of the 733 wavelengths it was trained on: all 185 of PRISMA's.



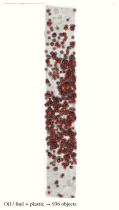
PRISMA



AVIRIS-NG



Tanager, man-made



AVIRIS-NG, oil + plastic

Deliverables

Per scene and selection

file	content
`hyper_anomaly_mask_<sel>.tif`	boolean mask, 0/1
`hyper_anomaly_pixels_<sel>.shp`	one point per pixel
`hyper_anomaly_objects_<sel>.shp`	one point per object
`hyper_anomaly_objects_<sel>.csv`	the same, as a table
`hyper_quicklook_<sel>.png`	RGB with the selection
`MANIFEST.json`	every choice in the chain
`<scene>_L2_SR.tif / .bsq`	surface reflectance

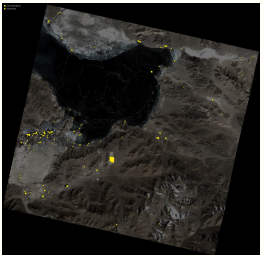
Mask GeoTIFF, PRISMA example

CRS	UTM zone 44N (EPSG:32644)
pixel	30 m, origin 375472.78 E, 3721282.25 N
values	8-bit, 0 or 1, no no-data value
man-made pack	110 objects, 865 pixels

Shapefile attributes

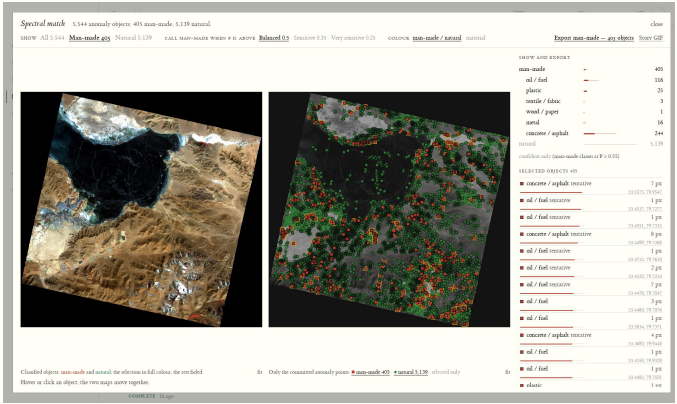
`lat, lon`	WGS84 degrees
`nm_call`	man-made or natural
`nm_mm`	object's P(man-made)
`nm_cls`	closest known class
`nm_p`	P(class | man-made)
`nm_conf`	1 confident, 0 tentative
`family`	ACE family match
`mat1, scr1`	best SAM library match, angle

The object table adds the object id, pixel count and centroid.



Quick-look: selected pixels in yellow, squares man-made, circles natural.

Product



Match screen: classified map and anomaly points, pan and zoom linked; man-made and natural, material classes, three sensitivity presets; the export button names the selection.

CLASS-DRIVEN EXPORT, END TO END

objects / mask pixels	PRISMA	Tanager
committed pixels	7,435	4,565
man-made	110 / 865	272 / 1,186
oil/fuel + plastic	26 / 598	247 / 1,124
natural	852 / 6,570	831 / 3,379
man-made, sensitive 0.35	240 / 1,672	424 / 2,073

Every pack checked: values 0/1, mask pixels and table rows equal the selection, shapefiles open in the scene CRS.

- Classes to leave out are chosen at scoring, from V2's class map.
- A story GIF shows the seven-step narrowing; it never enters a pack.
- Runs on an air-gapped GPU server from an offline update kit that keeps all data.

Summary

claim	evidence
Indradhanu V2 Heavy is our best detector	pAUC 0.189 vs FT04 0.170, +0.0196 [+0.0072, +0.0316]; 449 of 909 objects in the top 1 %
the model adds over no model	+0.047 pAUC over PCA-RX on the same scorer
HyperSL is not adopted	0.083 with its reconstruction, below 0.128 with none
the namer separates man-made from natural better than ACE	+0.20 / +0.17 / +0.14 recall at ACE's false-alarm rate, quarter-pixel fill
it names small targets better than ACE	class accuracy 0.59–0.69 against 0.24–0.41 at 10 % fill
L2 reflectance from L1, offline	0.0057 vs JPL on AVIRIS-NG; PRISMA 0.0099 vs ASI on unseen scenes
georeferenced, heatmap-free packs	the scene's own UTM grid; the 2.4-pixel PRISMA offset fixed
any sensor	PRISMA, EnMAP, AVIRIS-NG, Tanager, GHOSt and any GeoTIFF, ENVI or HDF5 cube

Limits and next

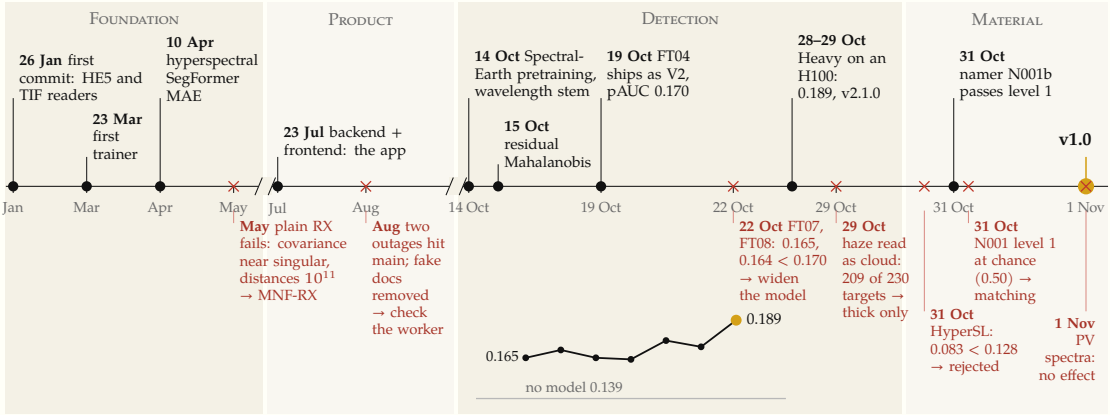
HONEST LIMITS

- **Open set.** An unseen material gets the nearest of seven classes; abstention catches 1–4 %.
- **Solar panels.** In-scene PV spectra added to the library changed nothing (P(man-made) 0.248 → 0.248); they need lab spectra or a retrain.
- **Calibration.** The 0.5 man-made call is not scene-calibrated; Tanager was not a training sensor.
- **Bench size.** Four labelled PRISMA scenes; resolution about $\pm$0.005–0.01 pAUC.
- **Cost.** Heavy is 1.7× FT04 on CPU: 454 vs 274 s per scene.
- **Thermal.** The chain is built (Chakshu) but no thermal scene has run on this stack.

NEXT

- Library augmentation for level 2 (run N003), swapped in without code changes.
- Field spectra for camouflage and military coatings; a glass and solar-panel class.
- A larger labelled bench, beyond PRISMA.

Our journey

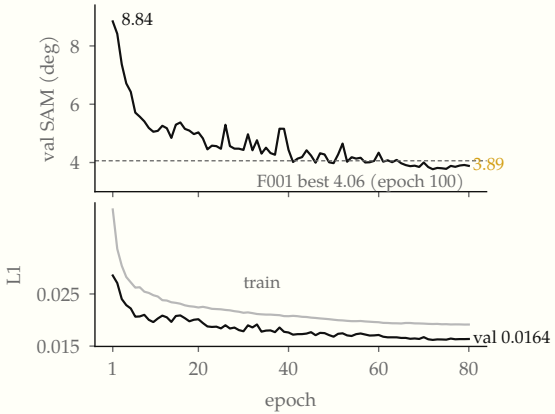


From a pipeline to a product: **any sensor in, georeferenced answers out.**

Allotrope v1.0 · 1 November 2026

A1 Heavy training

PRETRAINING ON SPECTRALEARTH, 202 BANDS



80 epochs; validation SAM 3.89°, below F001's best of 113 epochs.

RUN

hardware	one H100 80 GB, rented capacity block
throughput	369 s per epoch, 81.2 patches/s, 18.4 GB
data	3,200 of 3,419 shards, 30,000 patches per epoch
best checkpoint	epoch 72

PARAMETERS

part	FT04	Heavy
spectral stem	14,721	17,825
encoder	3,331,072	29,580,288
decoder	2,164,736	4,279,616
output layer + norm	4,321	6,465
segmentation head		1,431,120
served	5.9 M	**35,315,314**

PRETRAINED ONLY, NO FINE-TUNING

	pAUC	top 1 % / 0.1 %
F001, FT04 size	0.155	45.1 / 14.0 %
F002 Heavy	0.166	47.4 / 12.4 %

A2 Namer runs

run	change	result
N001	linear level-1 head	stopped after epoch 1: level 1 at chance, 0.49–0.52 (0.50 at full fill, where ACE gets 0.87)
N001b	level 1 by matching	passes on all three sensors, +0.20 / +0.17 / +0.14; **served, epoch 40**
N002	textile held out	open set fails: 0.46–0.47 called man-made
N003	library augmentation	results pending
PV library	50 in-scene PV spectra	no effect, 0.248 → 0.248; not shipped

ACE baseline, level 1 (man-made / natural correct)

fill	0.10	0.25	0.50	1.00
PRISMA	0.61 / 0.78	0.77 / 0.83	0.85 / 0.86	0.87 / 0.86
EnMAP	0.48 / 0.72	0.66 / 0.79	0.80 / 0.84	0.87 / 0.87
AVIRIS-NG	0.59 / 0.77	0.76 / 0.81	0.84 / 0.85	0.87 / 0.85

schedule

$$\lambda(s) = \min\Big(1, \tfrac{s+1}{500}\Big)\, \tfrac{1}{2}\Big(1 + \cos\big(\pi \min(1, \tfrac{s}{20000})\big)\Big)$$

AdamW; stem and its norm at 0.3× the base rate; weight decay 0.01; gradient clip 1. Each step: 3 sensors × 16 tiles × 512 pixels = 24,576 samples.

fill is log-uniform

$$P(a \le q) = \frac{\ln(q/a_{\min})}{\ln(1/a_{\min})}, \quad a_{\min} = 0.05$$

Most training mass sits at the small fills the detector finds.

chance levels

BCE at $p = \frac{1}{2}$: $\ln 2 = 0.693$; seven-class CE: $\ln 7 = 1.946$. N001 ended epoch 1 at BCE 0.684.

A3 HyperSL detail

per-scene min-max by hand

Scene 1: $p_1 = (0.10, 0.30)$, $p_2 = (0.20, 0.50)$, $p_3 = (0.30, 0.40)$ give $p_2' = (0.5, 1.0)$, $p_3' = (1.0, 0.5)$.
The same p_2 in scene 2 (spans 0.35, 0.40) gives $(0.4286, 0.75)$.
One bright anomaly $(0.60, 0.90)$ in scene 1 moves p_2' to $(0.2, 0.333)$.
Angle p_2, p_3: 15.07° before, 36.87° after. A library spectrum $(0.70, 0.20)$ in scene 1's frame is $(3.0, -0.5)$, outside $[0, 1]$.

cloud on EnMAP, recall and precision (%)

class	recall, HyperSL	recall, raw	precision, HyperSL	precision, raw
clear	89.0	89.4	88.1	90.8
cloud	**87.1**	80.4	77.8	**84.2**
cirrus	28.9	**67.9**	55.5	55.6
snow	86.5	**96.8**	93.5	91.0

HySU sub-pixel targets, object AUC

target side	0.25 m	0.5 m	1 m	2 m	3 m
ACE, library + field	**0.97**	**1.00**	1.00	1.00	1.00
ACE, library only	0.88	0.99	1.00	1.00	1.00
HyperSL anchored	0.68	0.77	1.00	1.00	1.00
HyperSL blind	0.72	0.62	0.80	0.88	0.92

the decision

need	HyperSL	Allotrope
same spectrum, same input	min-max per scene	fixed curve over wavelength
library spectra have an input	none	same stem
band lists with gaps	convolution over band index	40 nm wavelength bins
a pixel never rebuilds itself	rebuilds from its own bands	hides whole pixels

A4 Evaluation method

PARTIAL AUC, THE HEADLINE MEASURE

$$\mathrm{pAUC}_m = \frac{1}{m}\int_0^m \mathrm{TPR}(f)\,df = \frac{1}{P}\sum_p \frac{\max(0,\, m - \mathrm{FPR}(p))}{m}$$

Here $m = 0.01$, and $\mathrm{FPR}(p)$ is the share of background pixels scoring at or above target pixel p. Each object needs two numbers (the sum of its terms and its pixel count), so thousands of bootstrap draws are cheap.

THE OTHER MEASURES

- Objects caught at top q %: any pixel within 1 px of the top q % of scores. Chance: 10.2 % at 1 %, 1.1 % at 0.1 %.
- ROC-AUC, and the true-positive rate at a false-alarm rate of 10^{-3}.

THE BENCH

Four labelled PRISMA L2D scenes, 909 objects in three difficulty tiers; every method scored on the same validity and object maps.

PAIRED OBJECT BOOTSTRAP

- Resample the 909 objects with replacement, 2,000 draws; score both methods on each draw.
- Report $B - A$, its 2.5–97.5 % interval and $P(B > A)$; pairing removes shared object difficulty.
- Objects, not pixels, are the independent unit: $N_{\mathrm{eff}} = nm/(1 + (m-1)\rho)$ for n objects of m correlated pixels.
- Resolution on this bench: about $\pm$0.005–0.01 pAUC and $\pm$1–1.5 points of objects at the top 0.1 %.
- It does not cover checkpoint-to-checkpoint variation (FT09b's gain vanished five epochs later), so the release was checked at three neighbouring checkpoints.

A5 Deployment

atmospheric-correction container 1.3.0

build	builder, runtime without compilers, test stage
tests	81 passed, 54 % coverage (gate 30 %)
offline	runs with networking disabled; non-root; sees the GPU
size	22 GB image, 8.98 GB compressed, with a checksum
hosts	GPU, or CPU only

speed against accuracy, AVIRIS-NG window against JPL

superpixel size, per-pixel step	time	land MAE	median angle
50, analytical	40.8 min	0.0056	2.15°
200, analytical	17.8 min	0.0056	2.17°
200, empirical line	11.3 min	0.0056	2.17°
400, empirical line (default)	7.5 min	0.0057	2.19°

product update kit

- One archive per image (API, worker, frontend, database); IT scans each with Trivy.
- The kit writes a security folder: Trivy results for every severity, a high/critical table per image and a summary.
- Check, update and rollback commands; the kit replaces containers and never touches data volumes or settings.
- Rehearsed on a simulated copy of the air-gapped server.

worker cost, one PRISMA scene, 185 bands

	reconstruct	segment	score	total
Heavy, 2 threads	1629	185	4	1819 s
Heavy, 16 threads	402	47	4	454 s
FT04, 16 threads	236	32		274 s